

	Name
	Class
	R. No.

	• Mathematics
	• IIT JEE
	• Test 1 (July 11 , 2010)
	• Real Numbers
	• Maximum Marks 20
	• Time 20 minutes

Instructions

1. The test contains five subjective questions
2. The marking scheme will include marks for partially correct answers also.

Q1: Find all the rational values of x for which $\sqrt{49x^2 + 5x + 3}$ is a rational number.

6 Marks

Q2: Prove that for every positive rational number s satisfying the condition $s^2 > 7$, one can always find a smaller rational number $s - k$ ($k > 0$) for which $(s - k)^2 > 7$.

6 Marks

Q3: Prove that the sum $\sqrt{7} - \sqrt{2}$ is an irrational number.

2 Marks

Q4: Find the point $\sqrt{7} + \sqrt{22}$ on the number line using construction methods (Rough Sketch)

2 Marks

Q5: Prove that the sum and difference of a rational number(α) and an irrational number (β) is an irrational number. Also , the product and quotient is irrational provided $\alpha \neq 0$.

4 Marks